

SAKHI - YOUR AI MENTAL HEALTH CHATBOT

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Abstract

Mental health challenges among students and young adults are rising rapidly, yet social stigma and hesitation often prevent individuals from seeking help. Existing mental health applications rely heavily on formal, clinical, and robotic interactions, failing to provide the emotional warmth required by users in distress. This project introduces "Sakhi," an empathetic, AI-driven mental health companion designed specifically to bridge this gap. Utilizing the Llama-3.3-70b-versatile Large Language Model via the Groq API, Sakhi simulates culturally relatable Indian personas—such as a Mother, Father, or Best Friend—to foster a natural, human-like conversational environment. The system integrates quick distraction exercises and strict safety guardrails like the national Tele MANAS helpline. Through advanced prompt engineering and a seamless Streamlit-based interface, Sakhi demonstrates how artificial intelligence can move beyond simple query resolution to provide genuine psychological support and crisis de-escalation in everyday life.

Keywords: AI, Chatbot, Mental, Health, LLM

1. Problem Statement

In today's highly competitive and fast-paced socio-academic environment, students and young adults frequently experience extreme stress, loneliness, anxiety, and emotional exhaustion. Despite feeling overwhelmed by these pressures, a significant majority hesitate to discuss their mental health openly. This hesitation is largely driven by deep-rooted societal stigma, fear of judgment from peers, and a general lack of accessible, non-intimidating counseling resources.

While technology has introduced several mental health applications to the market, they often suffer from a critical usability flaw. Most of these platforms utilize rigid decision-tree bots or highly clinical language that feels overly formal and robotic. When a user is in a state of panic or deep sadness, they do not want to interact with a machine that simply lists out breathing exercises; they crave a human connection. Therefore, there is an urgent and critical need for a supportive digital platform that offers a friendly, culturally relatable, and empathetic environment. The goal is to design a system where users feel as though they are confiding in a close friend or family member, rather than navigating a sterile software interface.

2. Proposed System and Solution

To address the limitations of conventional clinical mental health apps, this project proposes "Sakhi," an innovative, AI-powered mental health companion. Sakhi is specifically tailored for the Indian cultural context, recognizing that emotional expression varies greatly across different demographics. Instead of adopting a standard, monotonous chatbot voice, Sakhi is engineered to simulate various relatable, real-life relationships based on the user's immediate emotional needs.

Users have the freedom to switch between multiple distinct avatars. For instance, they can select a protective 'Maa' (Mother) for unconditional warmth, a guiding 'Papa' (Father) for stoic motivation, a sarcastic yet supportive 'Bestie' (Friend) for a lighthearted distraction, or a 'Teacher' for philosophical guidance. This persona-driven approach ensures that conversations flow naturally, heavily utilizing localized language (Hinglish) and familiar cultural references (such as asking if the user has eaten).

Beyond conversation, Sakhi is equipped with practical wellness tools. The application interface includes a "Distraction Zone" featuring a 5-4-3-2-1 grounding exercise, lighthearted trivia, and simple puzzles to help users break out of anxiety loops. Crucially, the system incorporates a "Deep Talk Protocol" that detects severe distress and immediately shifts the tone to provide profound support, alongside embedded national helpline numbers (Tele MANAS 14416) for safety.

3. System Development Approach (Technology Used)

The architecture of Sakhi was designed to be lightweight, responsive, and highly intelligent. The development approach was divided into structured modules utilizing modern web and AI technologies:

- **Frontend Interface:** The primary user interface was developed using the *Streamlit* framework in Python. Streamlit was chosen for its rapid prototyping capabilities and its native ability to manage application state. Custom CSS was injected to create a soothing, Indian-inspired aesthetic with calming colors and clear typography. A secondary, static web version was also developed using standard HTML, CSS, and JavaScript for broader accessibility.
- **Backend Logic and State Management:** Python serves as the core programming language handling the application's backend logic. The system utilizes Streamlit's `session_state` to maintain a continuous chat history. This is vital for a mental health bot, as it allows the AI to remember the context of the conversation, making the interaction feel seamless.
- **Artificial Intelligence Engine:** The conversational intelligence is driven by the *Groq Cloud API*. We integrated the state-of-the-art `llama-3.3-70b-versatile` language model. Groq was selected over traditional APIs due to its exceptionally low latency, which mimics the real-time typing speed of a human being. The model's temperature was set dynamically to balance creativity with safe, coherent responses.
- **Security and Environment Variables:** To ensure the security of API credentials, the system employs the `dotenv` library to load sensitive keys from local environment variables, ensuring that unauthorized users cannot access the core AI engine directly.

4. Algorithm and Working Mechanism

The operational flow of the Sakhi chatbot relies on a continuous, state-aware feedback loop powered by advanced prompt engineering. The step-by-step algorithmic workflow is as follows:

1. **Initialization:** Upon launching the application, the system initializes the session state arrays and displays a welcoming message, establishing a safe space for the user.
2. **Persona Selection:** The user selects a desired avatar from the UI dropdown. The system dynamically updates the hidden "System Prompt" based on this selection.
3. **Input Capture:** The user types their feelings or queries into the input field. The system captures this text, renders it on the screen, and appends it to the active `messageHistory` array.

- 4. **Context Assembly:** The application constructs a JSON payload containing the heavily engineered system instructions (which restrict robotic language and enforce Hinglish), followed by the entire historical context of that specific session.
- 5. **API Transmission:** This structured payload is transmitted to the Groq API endpoint. The LLM processes the emotional context of the user’s latest input against the active persona constraints.
- 6. **Response Generation & Rendering:** The API returns the generated text. The system extracts this reply, updates the UI in real-time using Streamlit’s chat message containers, and saves the bot’s response back into the session state to maintain the conversational loop.

5. Results

Following the development and deployment phases, the Sakhi chatbot underwent practical testing with a sample group of 50 university students to evaluate its conversational efficacy, system latency, and empathetic accuracy. Quantitative and qualitative data were collected over a one-week testing period to ensure the system breaks the mold of traditional, rigid chatbots.

6. Performance Metrics

Table 1: Performance and User Engagement Metrics by Persona (Sample Size: 50 Users)

Persona Mode	Avg. Response Time (ms)	Empathy Rating (Out of 10)	Context Retention
Maa (Mother)	850	9.6	95
Papa (Father)	820	8.8	92
Bestie (Roast/Support)	810	9.4	96
Teacher (Guru)	890	9.1	94

Figure 1: Empathy Rating Comparison by Persona (Out of 10)



7. Observations

As observed in the performance metrics, the Groq API’s integration allowed the Llama-3.3-70b model to maintain exceptionally low latency, with an average response time of approximately 840 milliseconds across all modes. This near-instantaneous processing significantly enhanced the illusion of texting a real person. The ‘Maa’ and ‘Bestie’ personas scored the highest in Empathy Ratings (9.6 and 9.4 respectively), indicating that culturally nuanced prompt engineering strongly resonates with the target demographic.

Qualitative testing also demonstrated an impressive ability to context-switch. When a user expressed minor academic stress while in ‘Bestie’ mode, the bot responded with lighthearted

motivation and casual banter. Conversely, when the user expressed deep sadness in 'Maa' mode, the bot's tone shifted entirely to one of maternal concern, using comforting phrases and prioritizing emotional validation over factual advice. Furthermore, the UI's quick distraction buttons proved to be effective circuit-breakers for users experiencing sudden spikes in anxiety. The safety guardrails activated accurately during simulated crisis prompts, displaying the Tele MANAS details without fail.

8. Conclusion

The Sakhi project serves as a strong proof of concept that Artificial Intelligence can transcend its traditional roles of automation and data retrieval to offer genuine emotional companionship. By thoughtfully combining high-speed language models, rigorous prompt engineering, and deep cultural context, Sakhi provides an accessible, non-judgmental, and highly relatable digital support system. The performance metrics confirm that the system successfully mitigates the robotic feel of conventional mental health applications. Ultimately, Sakhi represents a meaningful step forward in leveraging technology to combat the growing epidemic of loneliness and daily stress among young adults in India.

9. Future Scope

While the current iteration of the Sakhi platform is robust and highly functional, it lays the groundwork for several advanced enhancements in the future:

- **Voice Integration:** Implementing speech-to-text and text-to-speech capabilities to allow users to have actual voice conversations with their chosen personas, further enhancing the illusion of human interaction.
- **Advanced Emotion Detection:** Integrating real-time sentiment analysis models that analyze user input not just for context, but for underlying emotional states, allowing the bot to automatically suggest switching to a more comforting persona if high distress is detected.
- **User Dashboards:** Adding a secure, authenticated backend database to track user moods over time, generating weekly mental health reports and visual charts for the user's personal reflection.
- **Vernacular Expansion:** Expanding the AI's prompt instructions to support pure regional Indian languages such as Bengali, Marathi, Tamil, and Telugu to reach rural and non-English/Hindi speaking demographics.
- **Professional Handoff:** Developing an API bridge that allows users to seamlessly connect their chat transcripts (with consent) to real, verified human therapists when the AI determines that professional intervention is necessary.

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